

Tehman Hassan

Final Year Computer Engineering Student
Department of Computer Engineering, University of Engineering
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2023-UET/CP-011
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Profile and Bio

Final-year Computer Engineering student at UET Taxila specializing in backend systems, distributed architecture, and AI-powered application development. Experienced in designing and deploying scalable REST APIs, real-time WebSocket systems, and microservices-based architectures using Python, Django, Fast API, and AWS — validated through a software engineering internship, freelance client deployments, and multiple hackathon-winning projects.

Education

Bachelor of Engineering in Computer Engineering

Aug 2023 - Present

University of Engineering and Technology Taxila, Pakistan

Currently in 6th Semester, having an updated CGPA of 3.38

Matric (Primary) and High School Intermediate

Apr 2017- Oct 2022

Cadet College Choa Saiden Shah Chakwal, Pakistan

A+ Grade. Activities and societies: Member of College Computing Club, Football Team, Debating Society, Appointment Holder
An institute that provided a disciplined and rigorous academic environment, focusing on character development and academic excellence. My time there involved a well-rounded curriculum and activities that shaped my foundation for further education.

Projects

Currently Leading Final Year Project with 3 fellow Students working in Agentic AI Niche

Personalized LMS System – 1st Runner Up in AI Hackathon

Project Link: <https://atomcamp.duckdns.org/>

- Built an adaptive learning platform that personalizes student paths in real time based on performance data
- Implemented dynamic quiz difficulty adjustment and instructor analytics dashboard
- Stack: Python, FastAPI, React, OpenAI API, PostgreSQL

Built “talentgraph.ai”, talent mapping in Global AI Hackathon

Project Link: <https://skillpath.duckdns.org/>

- Built a global skill-mapping platform that clusters talent by niche, location, and market trends
- Integrated AI-based job description parsing to match organizations with relevant talent pools
- Stack: Python, Django, React, OpenAI API, PostgreSQL

Full Stack Health Platform for Luther Health LTD

Project Link: <https://luther.health/>

- Developed a production-grade health management SaaS for a UK-based doctor with 15+ assessment funnels
- Reduced AI infrastructure costs by architecting an n8n workflow automation layer on AWS
- Stack: Python, Django, AWS EC2, n8n, OpenAI GPT-4, React, Stripe, PostgreSQL

Single Cycle Processor Datapath in Verilog HDL

Project Link: <https://github.com/Tehman700/Veriliog-HDL>

- Implemented a full single-cycle datapath supporting R, I, Load, Store, and Branch instruction types
- Stack: Verilog HDL, Vivado

Real-Time Asynchronous Chat App Using WebSocket's with End-to-End Encryption

Project Link: <https://github.com/Tehman700/Real-Time-Chat-App-2-Way-Communication-WebSockets>

- Built a full-stack async chat application with E2E encryption, OTP verification, and offline message handling
- Stack: Python, Django Channels, WebSockets, PostgreSQL, React

Potholes Complaint System (Winning Project) in OOP Course Exhibition

- Built an Android app using Azure Vision AI to verify pothole photo authenticity before filing complaints
- Designed a Spring Boot backend with PostgreSQL for complaint tracking and admin resolution management
- Stack: Java, Spring Boot, Azure Vision AI, PostgreSQL

AI Agent for Teachers (CLO & Bloom's Taxonomy Report Generator)

Project Link: https://github.com/Tehman700/Dr.Fawad_AI_Agent_Course

- Built a Java-based AI agent that analyses academic documents and auto-generates CLO-aligned reports
- Implemented tree data structures for document processing and vector ID storage for offline reliability
- Stack: Java, OpenAI GPT-4, PDF generation

Work Experience

Freelance Backend Software Developer

Aug 2025 – Jan 2026

Fiverr, Freelance Platform

- Built and deployed production-grade REST APIs using Python, Django REST Framework, and PostgreSQL, serving clients across multiple time zones
- Implemented real-time features using Django Channels and WebSockets for bi-directional communication in client applications
- Designed and maintained CI/CD pipelines on AWS EC2 with Nginx reverse proxy, reducing manual deployment effort and improving release reliability
- Achieved Level 1 Seller status within 2 months, maintaining 5-star client satisfaction through systematic debugging and error handling across AWS and server environments
- Integrated subscription billing flows and activity logging using Django signals for automated audit trails

Software Engineering Backend Intern

Jul 2025 – Sep 2025

Inara Technologies, F11 Markaz, Islamabad, Pakistan

- Architected a full-stack REST API from scratch using Django REST Framework, implementing JWT authentication, role-based access control, and custom exception handling
- Built a nested commenting system with likes, dislikes, and threaded replies across multiple content modules, supporting full CRUD operations
- Developed real-time two-way communication features using WebSockets and Django Channels for live user interaction
- Configured GitHub Actions CI/CD pipelines integrated with AWS deployment infrastructure, cutting manual release steps significantly

Skills

- **Languages:** Python, Java, JavaScript, Verilog, SQL
- **Frameworks:** Django REST, Fast-API, Node.js, Django Channels
- **Cloud & DevOps:** AWS EC2, S3, Nginx, GitHub Actions, Docker
- **Tools:** n8n, Postman, Git, WebSocket's
- **Databases:** PostgreSQL, MongoDB

Competitions

1st Runner Up in AI Hackathon at Aurex'26 at Bahria H11

Our team "Titans" won 2nd position in AI Hackathon where we built a personalized Learning Experience LMS using AI

Winner OOP Course Semester Project

Won the best Semester Project in my department among 150 students, along with certificates and prize money

Participation in Hackathon in Devsinc

This was a Speed Programming Competition in a Professional Software Company, tackling Data Structures, Greedy Algorithms, and DP Problems

Certificates

- API Testing Learning Path Certificate
- Postman API Fundamentals Student Expert
- Machine Learning with Python
- 100 Days of Code: The Complete Python Pro Bootcamp
- SQL Basic Hacker rank
- Transformers in LLMs Critical
- Thinking and Problem Solving (LinkedIn)
- Participation in Space Week Organized by NCGSA